Wireless Security Analyzer

ESP32-based portfolio project documentation

Project Overview

The supplied project is a modular ESP32 firmware application for wireless analysis. Its source is divided into application state, menu handling, OLED display handling, button input, Wi-Fi scanning, BLE scanning, and Wi-Fi channel analysis.

Key Features

• Wi-Fi network scanning with SSID, RSSI, channel, and security-type details
• BLE device scanning with device name, MAC address, and RSSI
• Wi-Fi channel analysis for channels 1–13
• SH1106 128×64 OLED interface over I²C
• Three-button navigation using UP, DOWN, and SELECT
• Modular C++/Arduino firmware structure

Hardware & Connections

Component	Connection / Role
ESP32	Main controller with built-in Wi-Fi and BLE
SH1106 OLED	I²C: SDA GPIO 21, SCL GPIO 22
UP button	GPIO 32, INPUT_PULLUP
DOWN button	GPIO 33, INPUT_PULLUP
SELECT button	GPIO 25, INPUT_PULLUP

Implementation Summary

The main program initializes the display and buttons, then enters a menu-driven state machine. The Wi-Fi scanner performs a station-mode network scan and provides a selectable list and detail screen. The BLE scanner uses NimBLE to collect up to 25 unique discovered devices. The channel analyzer counts detected Wi-Fi networks by channel and renders a simple bar view on the OLED.

Project Contribution

Designed and developed the project architecture, implemented the ESP32 firmware and wireless-analysis modules, integrated the OLED display and physical controls, developed the Wi-Fi/BLE scanning and channel-analysis functionality, and carried out testing and debugging of the system.

Project Status

The portfolio page marks this project as In Progress so that the website does not imply a finalized release beyond what is supported by the supplied source.

Responsible Use

This project should be used only for wireless networks and devices that the user owns or is explicitly authorized to assess. The portfolio presents the project for learning, diagnostics, and authorized security analysis.